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LIQUID APPLICATION APPARATUS

Abstract

A liquid application apparatus includes a body; a head in the body, the head to discharge a liquid from a nozzle; a carriage movable between a first position and a second position; a base; a plate slidable on a horizontal plane; a cap on the plate, the cap to cap the nozzle of the head; a first mover to move the carriage, to the first position above the cap, on the horizontal plane; a second mover to move the carriage to the cap in a vertical direction; a cleaner on the plate, the cleaner to discharge a cleaning liquid onto the nozzles at the second position; and a guide to slide the plate relative to the base on the horizontal plane to guide the cap to the carriage at the first position when the second mover moves the carriage to the cap in the vertical direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This patent application is based on and claims priority pursuant to 35 U.S.C. § 119(a) to Japanese Patent Application No. 2024-036997, filed on Mar. 11, 2024, in the Japan Patent Office, the entire disclosure of which is hereby incorporated by reference herein.

BACKGROUND

Technical Field

[0002] Embodiments of the present disclosure relate to a liquid application apparatus.

Related Art

[0003] In an ink jet apparatus or an image forming apparatus (or an apparatus that discharges liquid) that forms an image by discharging liquid such as ink, techniques for adjusting the position of a cleaning mechanism relative to a discharge head are known.

SUMMARY

[0004] An embodiment of the present disclosure provides liquid application apparatus including a body; a head in the body, the head to discharge a liquid from a nozzle; a carriage holding the head and movable between a first position and a second position; a base fixed to the body; a plate over the base and slidable on a horizontal plane parallel to a top face of the base; a cap on the plate, the cap to cap the nozzle of the head; a first mover to move the carriage, to the first position above the cap, on the horizontal plane; a second mover to move the carriage to the cap in a vertical direction orthogonal to the horizontal plane; a cleaner on the plate, the cleaner to discharge a cleaning liquid onto the nozzles of the head on the carriage at the second position to clean the nozzle; and a guide to slide the plate relative to the base on the horizontal plane to guide the cap to the carriage at the first position when the second mover moves the carriage to the cap in the vertical direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] A more complete appreciation of embodiments of the present disclosure and many of the attendant advantages and features thereof can be readily obtained and understood from the following detailed description with reference to the accompanying drawings, wherein:

[0006] FIG. **1** is a diagram illustrating an external appearance of an overall configuration of a liquid application apparatus;

[0007] FIG. **2** is a diagram illustrating a part of a liquid application apparatus, where a carriage can scan relative to a printing device;

[0008] FIG. **3** is a diagram illustrating a configuration of a moving mechanism of a carriage of a liquid application apparatus;

[0009] FIG. **4** is a diagram illustrating the operation of the liquid application apparatus when printing outside the scanning range of a carriage;

[0010] FIG. **5** is a diagram illustrating a liquid application apparatus being moved on a cart.

[0011] FIG. **6** is a diagram illustrating a printing device of a liquid application apparatus, placed on a printing surface after being unloaded from a cart;

[0012] FIG. **7** is a diagram of a hardware configuration of a liquid application apparatus;

[0013] FIGS. **8**A and **8**B are diagrams each illustrating a configuration of a maintenance mechanism of a liquid application apparatus;

[0014] FIG. **9** is an enlarged view of a stepped screw of a maintenance mechanism of a liquid

application apparatus;
[0015] FIGS. **10**A and **10**B are diagram illustrating an operation of inserting a pin of a carriage into a hole in a maintenance mechanism of a liquid application apparatus;
[0016] FIGS. **11**A, **11**B, and **11**C are diagrams for describing a cleaning operations of an ink discharge head of a liquid application apparatus;
[0017] FIGS. **12**A, **12**B, and **12**C are diagrams for describing a cleaning operations of an ink discharge head of a liquid application apparatus;
[0018] FIG. **13** is a diagram illustrating a configuration of a moving mechanism of a carriage of a liquid application apparatus;
[0019] FIGS. **14**A, **14**B, and **14**C are diagrams illustrating an operation of capping an ink discharge head in a liquid application apparatus; and
[0020] FIG. **15** is a diagram illustrating a configuration of a moving mechanism for a carriage in a liquid application apparatus.
[0021] The accompanying drawings are intended to depict embodiments of the present disclosure and should not be interpreted to limit the scope thereof. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. Also, identical or similar reference numerals designate identical or similar components throughout the several views.

DETAILED DESCRIPTION

[0022] In describing embodiments illustrated in the drawings, specific terminology is employed for the sake of clarity. However, the disclosure of this specification is not intended to be limited to the specific terminology so selected and it is to be understood that each specific element includes all technical equivalents that have a similar function, operate in a similar manner, and achieve a similar result.
[0023] Referring now to the drawings, embodiments of the present disclosure are described below. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.
[0024] An inkjet apparatus or an image forming apparatus (or an apparatus that discharges liquid) with such a cleaning mechanism includes a liquid discharge head having nozzles for discharging liquid on a nozzle surface, a wiping member for wiping the nozzle surface, a rotatable pressing member for pressing the wiping member against the nozzle surface, and a moving unit for performing a wiping operation of bringing the wiping member into contact with the nozzle surface and relatively moving the liquid discharge head and the wiping member.
[0025] Such an inkjet apparatus, however, faces the challenge of manually readjusting the position of the cleaning mechanism before printing when the relative position between the discharge head and the cleaning mechanism changes.
[0026] According to one aspect of the present disclosure, the position of the discharge head relative to the cleaning mechanism can be automatically adjusted to an appropriate position during cleaning, even if the relative position between the discharge head and the cleaning mechanism changes.
[0027] A liquid application apparatus according to an embodiment of the present disclosure will be described in detail with reference to the drawings. The present disclosure, however, is not limited to the following embodiment, and components of the following embodiment include components that may be easily conceived by those skilled in the art, components being substantially the same, and components being within equivalent ranges. Furthermore, various omissions, substitutions, changes, and combinations of the components can be made without departing from the gist of the following embodiment.

Overview of Overall Configuration and Operation of Liquid Application Apparatus

[0028] FIG. **1** is a diagram illustrating an external appearance of an overall configuration of a liquid application apparatus according to an embodiment of the present disclosure. FIG. **2** is a diagram illustrating a part of a liquid application apparatus according to an embodiment of the present disclosure, where a carriage can scan relative to a printing device.

[0029] FIG. **3** is a diagram illustrating a configuration of a moving mechanism of a carriage of a liquid application apparatus according to an embodiment of the present disclosure. The overall configuration and operation of a liquid application apparatus I will be described with reference to FIGS. **1** to **3**.

[0030] The liquid application apparatus **1** illustrated in FIG. **1** divides a wide liquid application area, such as a road surface, into multiple printing regions. The liquid application apparatus **1** then sequentially moves to each printing region, divides printing data for each printing region into multiple printing images, and discharges ink for printing. The ink is an example of liquid.

[0031] The road surface includes a road, a wall surface of a building. The road surface is simply referred to as a "printing surface". Further, "printing" refers to an operation of forming an image by applying or spraying ink onto the installation surface.

[0032] FIG. **2** indicates the internal structure of a printing device **10** with a front-side panel removed. The front side refers to the side closer to the viewer in the drawing sheet (FIG. **2**). As illustrated in FIG. **1**, the liquid application apparatus **1** includes a printing device **10** and a cart **20**.

[0033] The printing device **10** can be carried using the cart **20** and prints on the installation surface through the scanning of carriages **16** equipped with ink discharge heads **16***a*, as illustrated in FIG. **3**. As illustrated in FIGS. **1** and **2**, the printing device **10** includes a printing device body **11**, a controller **12**, an ink supply system **13**, stands **14**, and carriages **16**.

[0034] As illustrated in FIG. **3**, the printing device **10** includes a frame **11***a*, a main scanning guide **17**, a main scanning motor **17***a*, a sub-scanning guide **18**, a sub-scanning motor **18***a*, a timing belt **18***b*, and a height shifting mechanism **19**. The printing device **10** serves as a moving mechanism for scanning the carriages **16** equipped with the ink discharge heads **16***a*.

[0035] The moving mechanism is supported by four stands **14** mounted on a frame **11***a* that forms the peripheral edge of the bottom surface of the printing device body **11**. As illustrated in FIG. **3**, the printing device **10** includes a support member **11***b* and a maintenance mechanism **50**.

[0036] The printing device body **11** is a main body of the printing device **10**, within which the carriages **16** scan in the main scanning direction and the sub-scanning direction orthogonal to the main scanning direction. Various components, such as the controller **12** and the ink supply system **13** are mounted on the upper surface of the printing device body **11**. In FIG. **2**, four carriages **16** are moved to the four corners of the rectangular area that allows the scanning of the carriages **16** within the interior of the printing device body **11**.

[0037] The controller **12** controls the printing operation of the printing device **10**. Specifically, the controller **12** controls scanning of the carriage **16** in the main scanning direction, the sub-scanning direction, and a height direction, an ink discharging operation of the ink discharge heads **16***a* mounted on the carriages **16**, an ink supplying operation from the ink supply system **13** to the ink discharge heads **16***a*. The height direction is a vertical direction orthogonal to the main scanning direction and the sub-scanning direction, as well as the horizontal plane.

[0038] The controller **12** adjusts the height of each carriage **16** at multiple levels using the height shifting mechanism **19** to print images by discharging ink from the ink discharge heads **16***a*. To perform the maintenance operation of the maintenance mechanism **50** as described below, the controller **12** moves the carriages **16** in the height direction using the height shifting mechanism **19** so that the carriages **16** are positioned at a height for the maintenance operation.

[0039] The ink supply system **13** is mounted on the top surface of the printing device body **11** and supplies ink to the ink discharge heads **16***a* on the carriages **16**. Specifically, the ink supply system **13** includes a tank for storing ink, and supplies ink from the tank to the ink discharge heads **16***a* of the carriages **16** through a supply path such as a tube.

[0040] The stands **14** are support members that are respectively mounted at the four corners of the bottom surface of the rectangular parallelepiped printing device body **1**, supporting the printing device **10** by contacting the printing surface. The number of stands **14** is not limited to four, and

may be three, five, or more.
[0041] As illustrated in FIG. **3**, each carriage **16** is equipped with an ink discharge head **16***a* for ink discharge. The carriage **16** is scanned by the moving mechanism in the main scanning direction, the sub-scanning direction, and the height direction.
[0042] The scanning of the carriage **16** is controlled by the controller **12**. To print images using the ink discharge heads **16***a*, the carriages **16** are moved in the main scanning direction and the sub-scanning direction within a printing area AR as illustrated in FIG. **3**.
[0043] When not printing, the carriages **16** are moved to the maintenance mechanism **50** for the cleaning and capping operations, as illustrated in FIG. **3**.
[0044] Each ink discharge head **16***a* is an ink jet discharge head. Each ink discharge head **16***a* has multiple nozzles for discharging ink.
[0045] The frame **11***a* is a frame member that forms four sides of the bottom surface of the printing device body **11**. As illustrated in FIG. **3**, the stands **14** are mounted at the four corners of the bottom surface of the printing device body **11**, the four sides of which are formed by the frame **11***a*.
[0046] The main scanning guide **17** is a guide member that extends in the main scanning direction, as illustrated in FIG. **3** and supports the carriages **16** for sliding movement in the main scanning direction.
[0047] The main scanning motor **17***a* moves the height shifting mechanism **19** mounted with the carriage **16** back and forth along the main scanning guide **17** in the main scanning direction.
[0048] As illustrated in FIG. **3**, the sub-scanning guide **18** extends in the sub-scanning direction on the frame **11***a* and supports the main scanning guide **17**, allowing the main scanning guide **17** to slide in the sub-scanning direction. As illustrated in FIG. **3**, two sub-scanning guide **18** are placed on two opposing sides of the frame **11***a* that extend in the sub-scanning direction, to support the ends of the main scanning guide **17** extending in the main scanning direction.
[0049] The sub-scanning motor **18***a* moves the main scanning guide **17** back and forth in the sub-scanning direction along the sub-scanning guide **18**. In this case, driven by the rotation of the sub-scanning motor **18***a*, the timing belts **18***b*, which are wound around the driven pulleys, moves the main scanning guide **17** back and forth in the sub-scanning direction.
[0050] Thus, the carriage **16** mounted with the ink discharge head **16***a* moves freely in both the main scanning direction and the sub-scanning direction on the surface enclosed by the four sides of the frame **11***a*.
[0051] The height shifting mechanism **19** moves the carriages **16** back and forth in the height direction.
[0052] The support member **11***b* supports a base **50***a* of the maintenance mechanism **50**, which will be described later, and secures a plate **50***b* to the frame **11***b*.
[0053] The maintenance mechanism **50** is a mechanism for performing maintenance operations such as cleaning and capping the ink discharge head **16***a*. As illustrated in FIG. **3**, the maintaining mechanism **50** is supported by the support member **11***b*.
[0054] As illustrated in FIG. **3**, the maintenance mechanism **50** is placed outside the printing area AR for printing using the ink discharge head **16***a*.
[0055] As illustrated in FIG. **3**, the maintenance mechanism **50** includes a cap portion **51** (cap mechanism or simply referred to also as a cap) for capping the nozzles of the ink discharge head **16***a*, and a cleaner **52** (cleaning mechanism or simply referred to also as a clear) for cleaning the nozzles of the ink discharge head **16***a*. The details of the configuration of the maintaining mechanism **50** will be described later with reference to FIG. **9** and FIGS. **10**A and **10**B.
[0056] The cart **20** is a transport device used to lift the printing device **10** and from its bottom surface and transport it to a printing area for printing. As illustrated in FIG. **1**, the cart **20** includes a cart frame **21**, a lifting device **22**, a lifting device **23**, a rear wheel **24**, a front wheel **25**, and a handle portion **26**.
[0057] The cart frame **21** is a rectangular frame (or with a rectangular shape) that supports the

printing device **10** from the bottom surface during lifting and lowering.
[0058] The lifting device **22** supports the front portion of the printing device **10**, which is closer to the handle portion **26** and lifts and lowers the printing device **10**.
[0059] The lifting device **23** supports the rear portion of the printing device **10**, located opposite the handle portion **26** and lifts and lowers the printing device **10**.
[0060] The rear wheels **24** and the front wheels **25** allow the cart to move in both the front-rear direction and the right-left direction.
[0061] The handle portion **26** is attached to the front side of the cart **20** and is gripped by a user (or an operator). The user can freely move the cart **20** in the front-rear direction and the right-left direction by gripping the handle portion **26**.
Overview of Overall Operation of Liquid Application Apparatus
[0062] FIG. **4** is a diagram illustrating the operation of the liquid application apparatus when printing outside the scanning range of a carriage. An overview of the overall operation of a liquid application apparatus **1** will be described with reference to FIG. **4**.
[0063] As described above, the liquid application apparatus **1** is movable in four directions, i.e., forward, backward, leftward, and rightward (or in the front-rear direction and the right-left direction). In other words, a carriage **16** mounted with an ink discharge head **16***a* is freely movable within a plane enclosed by a frame **11***a* of the liquid application apparatus **1**.
[0064] In this configuration, the liquid application apparatus **1** operates as follows to allow the ink discharge head **16***a* to print on regions outside its scanning range on the printing surface. The liquid application apparatus **1** divides a liquid application area, which is an entire printing region of the printing surface, into multiple smaller printing regions; and also divides an entire printing data to be printed on the printing surface into multiple printing images, each corresponding to a specific printing region. Then, the liquid application apparatus **1** is moved backward, leftward, and rightward to each printing region over the liquid application, printing each printing image while joining the multiple printing images to minimize visible seams. Thus, the full image based on the printing data is completed.
Moving and Lifting/Lowering Operations of a Printing Device Using a Cart
[0065] FIG. **5** is a diagram illustrating a liquid application apparatus being moved on a cart. FIG. **6** is a diagram illustrating a printing device of a liquid application apparatus, placed on a printing surface after being unloaded from a cart. The moving and lifting/lowering operations of the printing device **10** in the liquid application apparatus **1** using a cart **20** is described with reference to FIGS. **5** and **6**.
[0066] In the liquid application apparatus **1** illustrated in FIG. **5**, the printing device **10** is lifted up by the cart **20** and moved, with the stands **14** separated from the printing surface. With the stands **14** of the printing device **10** being separated from the printing surface, the printing device **10** can be freely moved in both the front-rear direction and the right-left direction. This allows the printing device **10** to move to a desired printing region.
[0067] In the liquid application apparatus **1** illustrated in FIG. **6**, the printing device **10** is lowered by the cart **20** and supported by the stands **14** contacting the printing surface. With the printing device **10** supported by the stands **14** and fixed to the printing surface, printing can be performed by scanning the carriages **16** in the main scanning direction and the sub-scanning direction.
[0068] Although the liquid application apparatus **1** performs printing while moving relative to the printing surface, such as a road and a wall of a building, but it is not limited to this. The liquid application apparatus **1** can also be applied to typical inkjet image forming apparatuses and commercial printing machines that discharge ink from the ink discharge head **16***a* for printing.
Hardware Configuration Of Liquid Application Apparatus
[0069] FIG. **7** is a diagram of a hardware configuration of a liquid application apparatus **1**. The hardware configuration of the liquid application apparatus **1** is described below with reference to FIG. **7**.

[0070] As illustrated in FIG. **7**, a liquid application apparatus **1** includes a controller **12**, a carriage **16**, a main scanning movement mechanism **31**, a sub-scanning movement mechanism **32**, a height shifting mechanism **19**, a three-dimensional camera **33**, a two-dimensional camera **34**, a global navigation satellite (GNSS) receiver **35**, an operation panel **36**, and a cleaner **52**.

[0071] The controller **12** includes a central processing unit (CPU) **61**, a memory **62**, an interface (I/F**63**), and a unit control circuit **64**.

[0072] The CPU **61** is a calculation device that integrally controls the operation of the liquid application apparatus **1**. The CPU **61** is in data communication with the memory **62**, the I/F **63** and the unit control circuit **64** via the bus. The CPU **61** also controls the driving of the main scanning movement mechanism **31**, the sub-scanning movement mechanism **32**, the height shifting mechanism **19**, and an ink discharge head **16***a* via the unit control circuit **64**, and estimates the self-position of the liquid application apparatus **1** from a position measurement signal received by the GNSS receiver **35**.

[0073] The memory **62** is a storage medium, such as a read-only memory (ROM) or a random-access memory (RAM) that stores a program used to drive the CPU **61**.

[0074] The memory **62** is used as a work area of the CPU **61**.

[0075] The I/F **63** is a communication interface for connecting various external devices **40** such as tablet terminals, smartphones, personal computers (PCs), servers, and notebook PCs.

[0076] The unit control circuit **64** controls the operations of the main scanning movement mechanism **31**, the sub-scanning movement mechanism **32**, the height shifting mechanism **19**, and the ink discharge head **16***a* under the control of the CPU **61**.

[0077] A liquid application apparatus **1** includes circuitry (e.g., CPU **61**) controls the second mover (e.g., a height shifting mechanism **19**) to move the carriage, capped by the cap, away from the cap in the vertical direction; controls the first mover (e.g., the main scanning movement mechanism and the sub-scanning movement mechanism) to move the carriage from the first position to the second position above the cleaner; and controls the cleaner to discharge the cleaning liquid onto the nozzles of the head on the carriage at the second position.

[0078] The carriage **16** is mounted with an ink discharge head **16***a* that discharges ink onto the printing surface. The carriage **16** moves in the main scanning direction along the main scanning guide **17** illustrated in FIG. **3**, and the main scanning guide **17** moves in the sub-scanning direction, enabling the carriage **16** to move in the sub-scanning direction. The carriage **16** is also moved in the height direction by the height shifting mechanism **19**.

[0079] The three-dimensional camera **33** is a three-dimensional shape measurement device for circumference measurement. The three-dimensional camera **33** is mounted on the front portion of the printing device **10** and captures mages of the circumference of the liquid application apparatus **1**. The three-dimensional camera **33** transmits the captured images to the CPU **61** of the controller **12**. The three-dimensional camera **33** may be powered either by a battery mounted on the three-dimensional camera **33** or by an external power source for continuous operation.

[0080] The two-dimensional camera **34** is an imaging device that captures images of the printing surface and its surrounding area, including the printed image on the printing surface. Thus, the two-dimensional camera **34** faces downward. The two-dimensional camera **34** transmits the captured image to the CPU **61** of the controller **12**. The two-dimensional camera **34** may be powered either by a battery mounted on the three-dimensional camera **33** or by an external power source for continuous operation.

[0081] The circuitry (e.g., the CPU **61**) divides printing data into multiple printing images corresponding to printing regions on a printing surface; and controls the discharge head to discharge the liquid onto the printing regions on the printing surface.

[0082] The GNSS receiver **35** receives positioning signals from GNSSI from a positioning satellite based on a GNSS (for example, a global positioning system (GPS)) to measure a current position on the earth. The GNSS receiver **35** transmits the received positioning signals to the controller **12**.

[0083] The operation panel **36** allows input operations for the liquid application apparatus **1** and displays its processing results.

[0084] The main scanning movement mechanism **31** includes the main scanning guide **17** and the main scanning motor **17***a*. The main scanning movement mechanism **31** reciprocates the carriage **16** in the main scanning direction along the main scanning guide **17** under the control of the unit control circuit **64**.

[0085] The sub-scanning movement mechanism **32** includes the sub-scanning guide **18**, a sub-scanning motor **18***a*, and a timing belt **18***b*. The sub-scanning movement mechanism **32** reciprocates the main scanning guide **17** on the sub-scanning guide **18** in the sub-scanning direction orthogonal to the main scanning direction under the control of the unit control circuit **64**. This allows the carriage **16** supported by the main scanning guide **17** to reciprocates in the sub-scanning direction.

[0086] In other words, the main scanning movement mechanism **31** and the sub-scanning movement mechanism **32** move the carriage **16** on a horizontal plane. The main scanning movement mechanism **31** and the sub-scanning movement mechanism **32** correspond to a first movement mechanism or a first mover.

[0087] The height shifting mechanism **19** moves the carriages **16** back and forth in the height direction under the control of the unit control circuit **64**. The height shifting mechanism **19** corresponds to a second movement mechanism or a second mover.

[0088] The cleaner **52** is installed in the maintenance mechanism **50** as described above, and under the control of the CPU **61**, functions as a mechanism that sprays (or ejects) water or a cleaning solution containing a specified component upward to clean the nozzles of the ink discharge head **16***a*.

[0089] The hardware configuration of the liquid application apparatus **1** illustrated in FIG. **7** is merely an example, and the liquid application apparatus **1** may include additional components.

Configuration and Operation of Maintenance Mechanism

[0090] FIGS. **8**A and **8**B are diagrams each illustrating a configuration of a maintenance mechanism **50** of a liquid application apparatus **1**. FIG. **9** is an enlarged view of a stepped screw of a maintenance mechanism **50** of a liquid application apparatus **1**. FIGS. **10**A and **10**B are diagram illustrating an operation of inserting a pin of a carriage into a hole in a maintenance mechanism of a liquid application apparatus **1**. The configuration and operation of a maintenance mechanism **50** of the liquid application apparatus **1** will be described with reference to FIGS. **8** to **10**.

[0091] As illustrated in FIG. **8**A, the maintenance mechanism **50** includes a base **50***a*, a plate **50***b*, a cap portion **51**, a cleaner **52**, a pin **53**, stepped screws **54***a* and **54***b*, holes **55***a* and **55***b*, screw holes **56***a* and **56***b*, and plungers **57***a* and **57***b*.

[0092] The base **50***a* is a plate-shaped base member fixed to the frame **11***b* via the support member **11***a*. In other words, the base **50***a* is fixed to the printing device body **11**.

[0093] The plate **50***b* is a plate-shaped member positioned facing the upper surface of the base **50***a*, and has a plate surface substantially parallel to the upper surface.

[0094] The cap portion **51** is a cap mechanism for capping the nozzles of the ink discharge head **16***a* of the lowered carriage **16**. As illustrated in FIG. **8**A, the cap portion **51** is installed on the plate **50***b*.

[0095] The cleaner **52** sprays (or ejects) water or a cleaning solution containing a specified component upward to clean the nozzles of the ink discharge head **16***a*. The cleaner **52** sprays (or ejects) cleaning liquid with a specific impact force distribution. Since nozzle cleaning affects the discharging performance of the ink discharge head **16***a*, the ink discharge head **16***a* is to be positioned according to this impact force distribution during the cleaning of the cleaner **52**.

[0096] The pin **53** extends upward from the upper surface of the plate **50***b* and has a pointed tip. The pin **53** stands on the upper surface of the plate **50***b*. When the carriage **16** lowers to cap the nozzles with the cap portion **51**, the pin **53** fits into the hole **16***b* in the lower surface of the carriage

16 for alignment. The inner diameter of the hole **16***b* is either substantially equal to the outer diameter of the pin **53** or has a clearance with the pin **53** that allows for accurate positioning.
[0097] The pin **53** and the hole **16***b* in the carriage **16** are examples of a guide member.
[0098] The stepped screws **54***a* and **54***b* are stepped screw members for attaching the plate **50***b* to the base **50***a*. For the stepped screws **54***a* and **54***b*, the term "stepped screw **54**" is used to refer to any stepped screw or to collectively refer to the stepped screws **54***a* and **54***b*.
[0099] Regarding the holes **55***a* and **55***b* described later, the term "hole **55**" is used to refer to any hole or to collectively refer to the holes **55***a* and **55***b*. Regarding the screw holes **56***a* and **56***b* described later, the term "screw hole **56**" is used to refer to any hole or to collectively refer to the screw holes **56***a* and **56***b*.
[0100] As illustrated in FIG. **9**, the stepped screw **54** includes a head portion **71**, a stepped portion **72** having a diameter smaller than that of the head portion **71**, and a threaded portion **73** having a diameter smaller than that of the stepped portion **72**.
[0101] As illustrated in FIG. **9**, the stepped portion **72** and the threaded portion **73** are inserted into a hole **55** formed vertically in the plate **50***b*, with the lower surface of the head portion **71** contacting the upper surface of the plate **50***b*.
[0102] Further, the threaded portion **73** screws into a threaded screw hole **56** in the upper surface of the base **50***a*. As will be described later, the plungers **57***a* and **57***b* press the lower surface of the plate **50***b* upward, pressing the peripheral edge of the hole **55** against the lower surface of the head portion **71**. This secures the plate **50***b* to the base **50***a*.
[0103] However, the plate **50***b* is slidable on a plane (or the horizontal plane) parallel to the upper surface of the base **50***a* by sliding on or moving along the pressed plungers **57***a* and **57***b* and the lower surfaces of the head portion **71**. In this case, as illustrated in FIG. **9**, the plate **50***b* can slide within the diameter of the hole **55** that accommodates the stepped portion **72**.
[0104] The holes **55***a* and **55***b* are vertical through-holes in the plate **50***b*. As illustrated in FIGS. **8**A, **8**B, and **9**, the stepped portions **72** and the threaded portions **73** of the stepped screws **54***a* and **54***b* pass through the holes **55***a* and **55***b*, respectively.
[0105] The screw holes **56***a* and **56***b* formed in the upper surface of the base **50***a* and are threaded. As illustrated in FIGS. **9**, **9***a*, and **9** B, the threaded portions **73** of the stepped screws **54***a* and **54***b* are respectively screwed into the screw holes **56***a* and **56***b*.
[0106] The plungers **57***a* and **57***b* project upward from the base **50***a* and apply an upward force to the lower surface of the plate **50***b*.
[0107] FIG. **8**B illustrates the above-described maintenance mechanism **50** with the nozzles of the ink discharge head **16***a* capped with the cap portion **51**. When the nozzles are capped by the cap **51** after printing, the main scanning movement mechanism **31** and the sub-scanning movement mechanism **32** first position the carriage **16** at a position on a horizontal plane above the cap **51** of the maintenance mechanism **50**. This position on the horizontal plane is a capping position or a first position.
[0108] In a large printing device such as the liquid application apparatus **1**, twisting may occur in the printing device **10** due to vibrations during movement of the printing device **10**, impacts when the printing device **10** is placed on the printing surface by the stands **14**, vibrations caused by scanning of the carriage **16** in the main scanning direction, the sub-scanning direction, and the height direction, or position adjustment of the carriage **16** in the height direction. As a result, the cleaning position of the ink discharge head **16***a* of the carriage **16**, which is preset for cleaning by the cleaner **52**, may deviate from the originally intended appropriate cleaning position.
[0109] The appropriate cleaning position refers to the cleaning effect by the cleaner **52** on the ink discharge head **16***a* at that position being optimal or preferable. As a result, the capping position of the ink discharge head **16***a* with respect to the cap portion **51** fixed to the plate **50***b* as well as the cleaner **52** also deviates from the original appropriate position.
[0110] In this case, "appropriate capping position" refers to a state in which, when the ink

discharge head **16***a* is lowered from that position, the nozzles of the ink discharge head **16***a* are properly capped with the cap portion **51** so that the capping effect is optimal or suitable. Thus, when the carriage **16** is positioned at the capping position, as illustrated in FIG. **10**A, the axes of the hole **16***b* formed in the carriage **16** and the pin **53** standing on the upper surface of the base **50***a* are misaligned.

[0111] When the height shifting mechanism **19** moves the carriage **16** downward from this state as illustrated in FIG. **10**A, the tip portion (or end portion) of the pin **53** first enters the hole **16***b*, and the tapered portion of the tip portion is pushed by the peripheral edge of the hole **16***b* as the carriage **16** moves downward. As a result, as illustrated in FIG. **10**B, the pin **53** is inserted into and fitted into the hole **16***b* while the plate **50***b* slides with respect to the base **50***a*. Thus, the axis of the hole **16***b* and the axis of the pin **53** substantially align.

[0112] As a result, as illustrated in FIG. **8**B, the ink discharge head **16***a* is properly capped by the cap portion **51**. After the plate **50***b* slides, the position of the plate **50***b* relative to the base **50***a* is maintained by the pressure from the plungers **57***a* and **57***b*. Then, the capping position of the ink discharge head **16***a* is set to an appropriate position on the horizontal plane by the sliding of the plate **50***b*. Thus, the cleaning position of the ink discharge head **16***a* is also properly set. In other words, when the ink discharge head **16***a* is positioned at the cleaning position (i.e., a second position) on the horizontal plane, the ink discharge head **16***a* is correctly aligned with the cleaner **52** to ensure proper cleaning performance.

[0113] In some examples, a pin corresponding to the pin **53** of the maintenance mechanism **50** may be formed on the lower surface of the carriage **16**, and a hole corresponding to the hole **16***b* of the carriage **16** may be formed in the plate **50***b*.

Cleaning Operation of Ink Discharge Head

[0114] FIGS. **11**A, **11**B, **11**C, **12**A, **12**B, and **12**C are diagrams for describing the cleaning operations of an ink discharge head **16***a* of a liquid application apparatus **1**. The cleaning operation flows for the ink discharge head **16***a* of the liquid application apparatus **1** are described with reference to FIGS. **11**A, **11**B, and **11**C and FIGS. **12**A, **12**B, and **12**C.

[0115] First, after the printing of the ink discharge head **16***a* is completed, the main scanning movement mechanism **31** and the sub-scanning movement mechanism **32** move the carriage **16** to a predefined cleaning position as illustrated in FIG. **11**A. Then, as illustrated in FIG. **11**B, the cleaner **52** sprays (or ejects) water or a cleaning solution containing a specified component upward to clean the nozzles of the ink discharge head **16***a*. At this time, the cleaning position may shift from the originally intended position due to twisting in the printing device **10** as described above.

[0116] After the cleaning of the cleaner **52** is completed, the main scanning movement mechanism **31** and the sub-scanning movement mechanism **32** move the ink discharge head **16***a* to a predefined capping position as illustrated in FIG. **11**C. Then, as illustrated in FIG. **12**A, the height shifting mechanism **19** lowers the carriage **16** from the capping position, and the pin **53** is inserted and fitted to the hole **16***b* of the pin **53**. During this operation, the plate **50***b* slides relative to the base **50***a* so that the ink discharge head **16***a* can be properly capped by the cap portion **51** at an appropriate position. After the plate **50***b* slides, the position of the plate **50***b* relative to the base **50***a* is maintained by the pressure from the plungers **57***a* and **57***b*.

[0117] Then, when the next printing operation with the ink discharge head **16***a* is performed, the height shifting mechanism **19** raises the carriage **16** to release the capping state of the cap portion **51** as illustrated in FIG. **12**B. The main scanning movement mechanism **31** and the sub-scanning movement mechanism **32** move the carriage **16** to a predefined cleaning position as illustrated in FIG. **12**C. The cleaner **52** sprays water or a cleaning solution containing a specified component upward to the nozzles of the ink discharge head **16***a* of the carriage **16** at the appropriate cleaning position. At this time, the capping position of the ink discharge head **16***a* is set to an appropriate position by the sliding of the plate **50***b* as described above. Thus, the cleaning position of the ink discharge head **16***a* is also properly set. This results in a high cleaning effect on the nozzles of the ink discharge head **16***a*, maintaining the nozzles in proper condition. After that, printing with the ink discharge head **16***a* starts.

[0118] As described above, in the liquid application apparatus **1**, the carriage **16** is mounted with the ink discharge head **16***a*, and the main scanning movement mechanism **31** and the sub-scanning movement mechanism **32** move the carriage **16** on a horizontal plane.

[0119] The height shifting mechanism **19** moves the carriage **16** in the height direction, and the base **50***a* is fixed to the body of the liquid application apparatus **1**.

[0120] The plate **50***b* is placed between the carriage **16** and the base **50***a*, and slides horizontally (on the horizontal plane) relative to the upper surface of the base **50***a*. The cap portion **51** is mounted on the plate **50***b* and caps or covers the nozzles of the ink discharge head **16***a*.

[0121] The cleaner **52** is mounted on the plate **50***b* and sprays a cleaning liquid onto the nozzles. The main scanning movement mechanism **31** and the sub-scanning movement mechanism **32** moves the carriage **16** to the capping position to align the pin **53** with the hole **16***b*.

[0122] Upon lowering the carriage **16** from the capping position toward the cap portion **51** by the height shifting mechanism **19**, the plate **50***b* slides horizontally (or on the horizontal plane) on the base **50***a* to position the nozzles for their proper capping. Specifically, the plate **50***b* slides on a horizontal plane with respect to the base **50***a* to ensure proper capping of the nozzles with the cap portion **51**.

[0123] This enables automatic adjustment of the position of the ink discharge head **16***a*, enabling proper alignment with the cleaner **52** for its cleaning by the cleaner **52**, regardless of any relative positional changes. This results in a high cleaning effect on the nozzles of the ink discharge head **16***a*, maintaining the nozzles in proper condition.

First Modification

[0124] The liquid application apparatus **1** according to the first modification will be described, focusing on differences from the liquid application apparatus **1** according to the above-described embodiment. In the above-described embodiment, the pin **53** allows the plate **50***b* to slide relative to the base **50***a*. In the present modification, two pins are used to allow the plate **50***b* to slide relative to the base **50***a*. The overall configuration and hardware configuration of the liquid application apparatus **1** according to the present modification are the same as those described in the above embodiment.

Configuration and Operation of Maintenance Mechanism

[0125] FIG. **13** is a diagram illustrating a configuration of a moving mechanism of a carriage of a liquid application apparatus according to a first modification of an embodiment of the present disclosure. The configuration and operation of a maintenance mechanism **50-2** of a liquid application apparatus **1** will be described with reference to FIG. **13**.

[0126] As illustrated in FIG. **13**, the maintenance mechanism **50-2** includes a base **50***a*, a plate **50***b*, a cap portion **51**, a cleaner **52**, pins **53***a* and **53***b*, stepped screws **54***a* and **54***b*, holes **55***a* and **55***b*, screw holes **56***a* and **56***b*, and plungers **57***a* and **57***b*. In the maintaining mechanism **50-2**, the configuration, except for the pins **53***a* and **53***b*, is the same as that of the maintaining mechanism **50** in the above-described embodiment.

[0127] The pin **53***a* extends upward from the upper surface of the plate **50***b* and has a pointed tip. The pin **53***a* corresponds to the pin **53** of the maintaining mechanism **50** according to the above-described embodiment.

[0128] The pin **53***b* extends upward from the upper surface of the plate **50***b* and has a pointed tip. As illustrated in FIG. **13**, the pin **53***b* and the pin **53***a* have different lengths in the axial direction. Specifically, the pin **53***b* is shorter than the pin **53***a* in the axial direction. The pin **53***b* is positioned opposite the pin **53***a*, with the cap portion **51** in between. In other words, the pin **53***b* is positioned on the opposite side of the cap portion **51** from the pin **53***a*.

[0129] As illustrated in FIG. **13**, the carriage **16** has holes **16***b* and **16***c* in the lower surface thereof. The hole **16***c* is elongated, with its major axis in the direction of the line connecting the centers of the holes **16***b* and **16***c*, as illustrated in FIG. **13**. The major axis of the elongated hole **16***c* extends along the line connecting the centers of the holes **16***b* and **16***c*. The inner diameter of the hole **16***b* is either substantially equal to the outer diameter of the pin **53***a* or has a clearance with the pin **53***a* that allows for accurate positioning as described above.

[0130] The pins **53***a* and **53***b*and the hole **16***b* in the carriage **16** are examples of a guide.

[0131] FIG. **13** illustrates the above-described maintenance mechanism **50-2** with the nozzles of the ink discharge head **16***a* capped with the cap portion **51**. When the cap portion **51** is used for capping after printing is finished, the main scanning movement mechanism **31** and the sub-scanning movement mechanism **32** first position the carriage **16** at a position on the horizontal plane (i.e., the capping position) above the cap portion **51** of the maintenance mechanism **50**. When the height shifting mechanism **19** moves the carriage **16** downward from this state as illustrated in FIG. **13**, the tip portion of the pin **53***a* longer than the pin **53***b* first enters the hole **16***b*, and the tapered portion of the tip portion is pushed by the peripheral edge of the hole **16***b* as the carriage **16** moves downward. Thus, the plate **50***b* slides with respect to the base **50***a*. Further, when the tip portion of the pin **53***b* enters the elongated hole **16***c*, the movement of the pin **53***b* is restricted along its minor axis. The pin **53***a* is inserted into and fitted into the hole **16***b*, and thus, the axis of the hole **16***b* and the axis of the pin **53***a* substantially align. Further, the direction of the line connecting the centers of the holes **16***b* and **16***c* coincides with the direction connecting the axes of the pins **53***a* and **53***b*. The ink discharge head **16***a* is properly capped by the cap portion **51**. After the plate **50***b* slides, the position of the plate **50***b* relative to the base **50***a* is maintained by the pressure from the plungers **57***a* and **57***b*. Then, the position at which the cap portion **51** caps the ink discharge head **16***a* is set to an appropriate position on the horizontal plane by the sliding of the plate **50***b*. Thus, the cleaning position of the ink discharge head **16***a* is also properly set on the horizontal plane. In other words, when the ink discharge head **16***a* is positioned at the cleaning position, the ink discharge head **16***a* is correctly aligned with the cleaner **52** to ensure proper cleaning performance.

[0132] A liquid application apparatus includes a body; a head (e.g., ink discharge head **16***a*) in the body, the head to discharge a liquid from a nozzle; a carriage (e.g., a carriage **16**) holding the head and movable between a first position and a second position; a base (e.g., a base **50***a*) fixed to the body; a plate (e.g., a plate **50***b*) over the base and slidable on a horizontal plane parallel to a top face of the base; a cap (e.g., a cap portion **51**) on the plate, the cap to cap the nozzle of the head; a first mover (e.g., a main scanning movement mechanism **31** and a sub-scanning movement mechanism **32**) to move the carriage, to the first position above the cap, on the horizontal plane; a second mover (e.g., a height shifting mechanism) to move the carriage to the cap in a vertical direction orthogonal to the horizontal plane; a cleaner (e.g., a cleaner **52**) on the plate; and a guide (e.g., a pin **53** and a hole **16**). The cleaner discharges a cleaning liquid onto the nozzles of the head on the carriage at the second position to clean the nozzle; and a guide to slide the plate relative to the base on the horizontal plane to guide the cap to the carriage at the first position when the second mover moves the carriage to the cap in the vertical direction.

[0133] In some examples, pins corresponding to the pins **53***a* and **53***b* of the maintenance mechanism **50-2** may be formed on the lower surface of the carriage **16**, and holes corresponding to the holes **16***b* and **16***c* of the carriage **16** may be formed in the plate **50***b*.

Capping Operation of Ink Discharge Head

[0134] FIGS. **14**A, **14**B, and **14**C are diagrams illustrating an operation of capping an ink discharge head **16***a* in a liquid application apparatus **1**, according to a first modification of the above-described embodiment of the present disclosure. The capping operation of the ink discharge head **16***a* during cleaning in the liquid application apparatus **1** of the present modification is described below with reference to FIGS. **14**A, **14**B, and **14**C.

[0135] After the cleaning of the cleaner **52** is completed, the main scanning movement mechanism **31** and the sub-scanning movement mechanism **32** move the ink discharge head **16***a* to a predefined capping position as illustrated in FIG. **14**A. Then, as illustrated in FIG. **14**B, the height shifting mechanism **19** lowers the carriage **16** from the capping position, and as described above, the plate **50***b*slides relative to the base **50***a*. Then, as illustrated in FIG. **14**C, the pin **53***a* is inserted and fitted to the hole **16***b*, and the pin **53***b* is inserted and fitted to the hole **16***c*. The ink discharge head **16***a* is properly capped by the cap portion **51**. After the plate **50***b* slides, the position of the plate **50***b* relative to the base **50***a* is maintained by the pressure from the plungers **57***a* and **57***b*.

[0136] This configuration of the present modification enables automatic adjustment of the position of the ink discharge head **16***a*, enabling proper alignment with the cleaner **52** for its cleaning by the cleaner **52**, regardless of any relative positional changes. This results in a high cleaning effect on the nozzles of the ink discharge head **16***a*, maintaining the nozzles in proper condition.

Second Modification

[0137] The liquid application apparatus **1** according to a second modification will be described, focusing on differences from the liquid application apparatus **1** according to the above-described embodiment. In the above-described embodiment, the pin **53** allows the plate **50***b* to slide relative to the base **50***a*. In the present modification, a guide member is used to allow the plate **50***b* to slide relative to the base **50***a*. The overall configuration and hardware configuration of the liquid application apparatus **1** according to the present modification are the same as those described in the above embodiment.

Configuration and Operation of Maintenance Mechanism

[0138] FIG. **15** is a diagram illustrating a configuration of a moving mechanism of a carriage of a liquid application apparatus according to a second modification of an embodiment of the present disclosure. The configuration and operation of a maintenance mechanism **50-3** of a liquid application apparatus **1** will be described with reference to FIG. **15**.

[0139] As illustrated in FIG. **15**, the maintenance mechanism **50-3** includes a base **50***a*, a plate **50***b*, a cap portion **51**, a cleaner **52**, stepped screws **54***a* and **54***b*, holes **55***a* and **55***b*, screw holes **56***a* and **56***b*, plungers **57***a* and **57***b*, and guide members **58***a* and **58***b*. In the maintaining mechanism **50-3**, the configuration, except for the pins **53***a* and **53***b*, is the same as that of the maintaining mechanism **50** in the above-described embodiment. The maintenance mechanism **50-3** includes guide members **58***a* and **58***b* in place of the pin **53** in the maintenance mechanism **50**. The carriage **16** does not have the hole **16***b* as in the above-described embodiment.

[0140] The guide members **58***a* and **58***b* extend upward from the opposing positions on the upper surface of the plate **50***b*, with the cap portion **51** interposed therebetween. The guide members **58***a* and **58***b* are tapered, with the distance between their opposing surfaces decreasing downward, as illustrated in FIG. **15**. The guide members **58***a* and **58***b* are examples of a guide.

[0141] FIG. **15** illustrates the above-described maintenance mechanism **50-3** with the nozzles of the ink discharge head **16***a* capped with the cap portion **51**. When the cap portion **51** is used for capping after printing, the main scanning movement mechanism **31** and the sub-scanning movement mechanism **32** first position the carriage **16** at a position (i.e., the capping position) above the cap portion **51** of the maintenance mechanism **50**. When the height shifting mechanism **19** lowers the carriage **16** from this state, the peripheral edge of the lower surface of the carriage **16** contacts one of the opposing surfaces of the guide members **58***a* and **58***b*. The carriage **16** is then pushed by the one contacted surface toward the other opposing surface. Thus, the plate **50***b* slides with respect to the base **50***a*. The ink discharge head **16***a* is properly capped by the cap portion **51**. After the plate **50***b* slides, the position of the plate **50***b* relative to the base **50***a* is maintained by the pressure from the plungers **57***a* and **57***b*.

[0142] The liquid application apparatus **1** further includes two guides (e.g., **58***a*, **58***b*), the two guides opposing on the plate, with the cap interposed therebetween. The two guides are tapered with a distance between opposing surfaces of the two guides decreasing in a direction toward the plate. When the second mover (e.g., the height shifting mechanism) moves the carriage to the cap in the vertical direction, a peripheral edge of the carriage contacts and presses one of the opposing surfaces to slide the plate relative to the base on the horizontal plane to guide the cap to the carriage at the first position.

[0143] Then, the capping position of the ink discharge head **16***a* is set to an appropriate position by the sliding of the plate **50***b*. Thus, the cleaning position of the ink discharge head **16***a* is also properly set. In other words, when the ink discharge head **16***a* is positioned at the cleaning position, the ink discharge head **16***a* is correctly aligned with the cleaner **52** to ensure proper cleaning performance. This results in a high cleaning effect on the nozzles of the ink discharge head **16***a*, maintaining the nozzles in proper condition.

[0144] In the liquid application apparatus **1** of the present modification, no holes are formed in the carriages **16**, whereas the plate **50***b* includes the guide members **58***a* and **58***b* in place of the pin **53**. This simple configuration achieves the above-described effects.

[0145] Aspects of the present disclosure are as follows.

[0146] Aspect 1

[0147] A liquid application apparatus includes a body; a head in the body, the head to discharge a liquid from a nozzle; a carriage holding the head and movable between a first position and a second position; a base fixed to the body; a plate over the base and slidable on a horizontal plane parallel to a top face of the base; a cap on the plate, the cap to cap the nozzle of the head; a first mover to move the carriage, to the first position above the cap, on the horizontal plane; a second mover to move the carriage to the cap in a vertical direction orthogonal to the horizontal plane; a cleaner on the plate, the cleaner to discharge a cleaning liquid onto the nozzles of the head on the carriage at the second position to clean the nozzle; and a guide to slide the plate relative to the base on the horizontal plane to guide the cap to the carriage at the first position when the second mover moves the carriage to the cap in the vertical direction.

[0148] Aspect 2

[0149] The liquid application apparatus according to Aspect 1, further includes circuitry (e.g., CPU **61**) controls the second mover to move the carriage, capped by the cap, away from the cap in the vertical direction; controls the first mover to move the carriage from the first position to the second position above the cleaner; and controls the cleaner to discharge the cleaning liquid onto the nozzle of the head on the carriage at the second position.

[0150] Aspect 3

[0151] In the liquid application apparatus according to Aspect 1 or 2, the guide includes a pin having a tip portion, the pin standing on the plate; and a hole in the carriage, the hole into which the pin is insertable.

[0152] Aspect 4

[0153] In the liquid application apparatus according to Aspect 1 or 1, the guide includes a pin having a tip portion, the pin standing on a face of the carriage; and a hole in the plate, the hole into which the pin is insertable.

[0154] Aspect 5

[0155] In the liquid application apparatus according to Aspect 1 or 2, the guide includes two pins each having a tip portion, the two pins standing on the plate; and two holes in the carriage, the two holes into which the two pins are respectively insertable.

[0156] Aspect 6

[0157] In the liquid application apparatus according to Aspect 1 or 2, the guide includes two pins each having a tip portion, the two pins standing on the carriage; and two holes in the plate, the two holes into which the two pins are respectively insertable.

[0158] Aspect 7

[0159] In the liquid application apparatus according to Aspect 5 or 6, the two holes have a circular hole having a first center; and an elongated hole having a second center along a major axis. The major axis coincides with a line connecting the first center and the second center.

[0160] Aspect 8

[0161] In the liquid application apparatus according to Aspect 5 or 6, the two pins have different length in an axial direction thereof.
[0162] Aspect 9
[0163] The liquid application apparatus according to Aspect 1 or 2, further includes two guides including the guide, the two guides opposing on the plate, with the cap interposed therebetween. The two guides are tapered with a distance between opposing surfaces of the two guides decreasing in a direction toward the plate. When second mover moves the carriage to the cap in the vertical direction, a peripheral edge of the carriage contacts and presses one of the opposing surfaces to slide the plate relative to the base on the horizontal plane to guide the cap to the carriage at the first position.
[0164] Aspect 10
[0165] The liquid application apparatus according to any one of Aspects 1 to 9, further includes circuitry (e.g., the CPU **61**) divides printing data into multiple printing images corresponding to printing regions on a printing surface; and controls the discharge head to discharge the liquid onto the printing regions on the printing surface.
[0166] Aspect 11
[0167] In the liquid application apparatus according to Aspect 1, the first mover moves the carriage between the first position and the second position in a main scanning direction on the horizontal plane; and moves the carriage in a sub-scanning direction orthogonal to the main scanning direction on the horizontal plane.
[0168] The above-described embodiments are illustrative and do not limit the present invention. Thus, numerous additional modifications and variations are possible in light of the above teachings. For example, elements and/or features of different illustrative embodiments may be combined with each other and/or substituted for each other within the scope of the present invention.
[0169] The functionality of the elements disclosed herein may be implemented using circuitry or processing circuitry which includes general purpose processors, special purpose processors, integrated circuits, application-specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs), and/or combinations thereof which are configured or programmed, using one or more programs stored in one or more memories, to perform the disclosed functionality. Processors are considered processing circuitry or circuitry as they include transistors and other circuitry therein. In the disclosure, the circuitry, units, or means are hardware that carry out or are programmed to perform the recited functionality. The hardware may be any hardware disclosed herein which is programmed or configured to carry out the recited functionality.
[0170] There is a memory that stores a computer program which includes computer instructions. These computer instructions provide the logic and routines that enable the hardware (e.g., processing circuitry or circuitry) to perform the method disclosed herein. This computer program can be implemented in known formats as a computer-readable storage medium, a computer program product, a memory device, a record medium such as a CD-ROM or DVD, and/or the memory of an FPGA or ASIC.

Claims

1. A liquid application apparatus comprising: a body; a head in the body, the head to discharge a liquid from a nozzle; a carriage holding the head and movable between a first position and a second position; a base fixed to the body; a plate over the base and slidable on a horizontal plane parallel to a top face of the base; a cap on the plate, the cap to cap the nozzle of the head; a first mover to move the carriage, to the first position above the cap, on the horizontal plane; a second mover to move the carriage to the cap in a vertical direction orthogonal to the horizontal plane; a cleaner on the plate, the cleaner to discharge a cleaning liquid onto the nozzles of the head on the carriage at the second position to clean the nozzle; and a guide to slide the plate relative to the base on the horizontal plane to guide the cap to the carriage at the first position when the second mover moves the carriage to the cap in the vertical direction.

2. The liquid application apparatus according to claim 1, further comprising circuitry configured to: control the second mover to move the carriage, capped by the cap, away from the cap in the vertical direction; control the first mover to move the carriage from the first position to the second position above the cleaner; and control the cleaner to discharge the cleaning liquid onto the nozzle of the head on the carriage at the second position.

3. The liquid application apparatus according to claim 1, wherein the guide includes: a pin having a tip portion, the pin standing on the plate; and a hole in the carriage, the hole into which the pin is insertable.

4. The liquid application apparatus according to claim 1, wherein the guide includes: a pin having a tip portion, the pin standing on a face of the carriage; and a hole in the plate, the hole into which the pin is insertable.

5. The liquid application apparatus according to claim 1, wherein the guide includes: two pins each having a tip portion, the two pins standing on the plate; and two holes in the carriage, the two holes into which the two pins are respectively insertable.

6. The liquid application apparatus according to claim 1, wherein the guide includes: two pins each having a tip portion, the two pins standing on the carriage; and two holes in the plate, the two holes into which the two pins are respectively insertable.

7. The liquid application apparatus according to claim 5, wherein the two holes have: a circular hole having a first center; and an elongated hole having a second center along a major axis, and the major axis coincides with a line connecting the first center and the second center.

8. The liquid application apparatus according to claim 5, wherein the two pins have different length in an axial direction thereof.

9. The liquid application apparatus according to claim 1, further comprising two guides including the guide, the two guides opposing on the plate, with the cap interposed therebetween, the two guides being tapered with a distance between opposing surfaces of the two guides decreasing in a direction toward the plate, wherein when the second mover moves the carriage to the cap in the vertical direction, a peripheral edge of the carriage contacts and presses one of the opposing surfaces to slide the plate relative to the base on the horizontal plane to guide the cap to the carriage at the first position.

10. The liquid application apparatus according to claim 1, further comprising circuitry configured to: divide printing data into multiple printing images corresponding to printing regions on a printing surface; and control the head to discharge the liquid onto the printing regions on the printing surface.

11. The liquid application apparatus according to claim 1, wherein the first mover: moves the carriage between the first position and the second position in a main scanning direction on the horizontal plane; and moves the carriage in a sub-scanning direction orthogonal to the main scanning direction on the horizontal plane.